# KARBON KISLOTALAR

Karboksil guruhli birikmalar · R–COOH

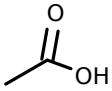
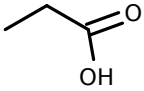
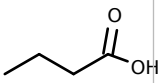
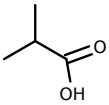
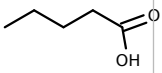
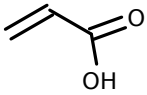
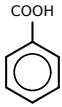
Student book – nazariy qo‘llanma

Molekulasida **karboksil guruhi (–COOH)** bo‘lgan organik birikmalar **karbon kislotalar** deyiladi. Karboksil guruhi **karbonil (>C=O)** va **gidroksil (–OH)** guruhlarining birikishidan hosil bo‘lgan. To‘yingan bir negizli kislotalarning umumiy formulasi $\mathbf{C_nH_{2n+1}COOH}$ yoki $\mathbf{C_nH_{2n}O_2}$ (n – jami uglerod soni); $M_r$ = 14n + 32.

**Muhim:** bir xil $C_nH_{2n}O_2$ tarkibli karbon kislota va **murakkab efir** – sinflararo izomerlar. Skelet formulalarda –COOH guruhi tarkibidagi O va OH ko‘rsatilgan.

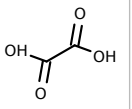
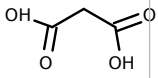
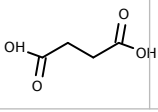
# 1. Tasnifi va nomlanishi

## A) Bir negizli kislotalar

| Formula | Skelet | Sistematik nomi | Tarixiy nomi |
|---|---|---|---|
| HCOOH | | metan kislota | chumoli kislota |
| $CH_3COOH$ |  | etan kislota | sirka kislota |
| $C_2H_5COOH$ |  | propan kislota | propion kislota |
| $C_3H_7COOH$ |  | butan kislota | moy kislota |
| $(CH_3)_2CH–COOH$ |  | 2-metilpropan kislota | izomoy kislota |
| $C_4H_9COOH$ |  | pentan kislota | valerian kislota |
| $CH_2=CH–COOH$ |  | propen kislota | akril kislota |
| $C_6H_5COOH$ |  | benzoy kislota | benzol karbon kislota |

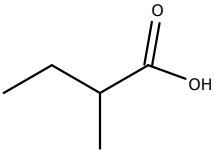
Yuqori yog‘ kislotalari: **$C_{15}H_{31}COOH$** – palmitin, **$C_{17}H_{35}COOH$** – stearin (to‘yingan), **$C_{17}H_{33}COOH$** – olein (to‘yinmagan, bitta C=C).

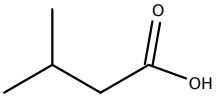
### B) Ikki negizli kislotalar

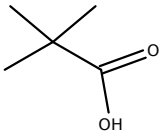
| Formula | Skelet | Sistematik nomi | Tarixiy nomi |
|---|---|---|---|
| HOOC–COOH |  | etandi kislota | oksalat (shavel) kislota |
| HOOC–$CH_2$–COOH |  | propandi kislota | malon kislota |
| HOOC–$(CH_2)_2$–COOH |  | butandi kislota | yantar kislota |

### C) Nomlash qoidalari

- Alkan nomiga **«kislota»** qo‘shiladi (etan → etan kislota). Zanjir **–COOH uglerodidan** raqamlanadi; u doim **1-raqamni** oladi.
- Eng uzun zanjir karboksil uglerodni o‘z ichiga oladi; radikallar raqamlari bilan alifbo tartibida yoziladi (2-metilbutan kislota).
- Ikki negizli kislotalarda **«di»** qo‘shimchasi ishlatiladi: **propandi kislota**.
- Tarixiy nomlar kislotaning topilgan manbaidan olingan: chumolidan – **chumoli**, sirkadan – **sirka**, valeriana o‘simligidan – **valerian**.


2-metilbutan kislota


3-metilbutan kislota


2,2-dimetilpropan kislota

## 2. Tuzilishi va kislotali xossasi

- Karboksil guruhdagi uglerod **$sp^2$-gibridlangan**; guruh **yassi**. Karbonil guruhning elektron tortuvchi ta’siri tufayli **O–H bog‘i kuchli qutblangan**.
- Gidroksil kislorodining juft elektronlari karbonilning π-bog‘i bilan qoplanadi (p,π-qo‘shilish): O–H dagi vodorod **oson ajraladi**, kislota suvli eritmada dissotsiatsiyalanadi: **$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$**.
- Karbon kislotalar **spirt va fenollardan kuchli** kislotalar, lekin **kuchsiz elektrolitlar** (mineral kislotalardan kuchsiz).
- Radikal uzayishi bilan kislotalik **kamayadi**: HCOOH eng kuchli. Elektronegativ atom (Cl) kislotalikni **oshiradi**: $ClCH_2COOH > CH_3COOH$.

- Molekulalar orasida **vodorod bog‘lar** kuchli – kislotalar **dimer** holda bo‘ladi, shuning uchun qaynash harorati mos spirtlardan **yuqori**.

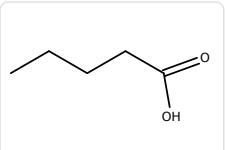
| Modda | Formula | $t_{qayn}$, °C |
|---|---|---|
| propanol-1 | $C_3H_7OH$ | 97 |
| chumoli kislota | HCOOH | 101 |
| sirka kislota | $CH_3COOH$ | 118 |
| propion kislota | $C_2H_5COOH$ | 141 |

Kislotalik qatori: $C_2H_5OH < H_2O < C_6H_5OH < H_2CO_3 < CH_3COOH < HCOOH$

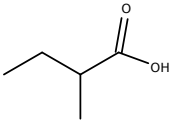
## 3. Izomeriyasi

Karbon kislotalarda **uglerod skeleti** va **sinflararo** (murakkab efirlar bilan) izomeriya uchraydi.

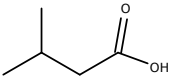
**$C_5H_{10}O_2$** tarkibli to‘yingan kislotalar – **4 ta** izomer:



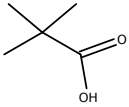
pentan kislota



2-metilbutan kislota



3-metilbutan kislota



2,2-dimetilpropan kislota

**$C_4H_8O_2$**: kislotalar – 2 ta (butan va 2-metilpropan kislota); murakkab efirlar – 4 ta (metilpropionat, etilasetat, propilformiat, izopropilformiat).

## 4. Fizik xossalari

- $C_1$–$C_3$ kislotalar – **o‘tkir hidli suyuqliklar**, suv bilan **istalgan nisbatda** aralashadi; $C_4$–$C_9$ – yoqimsiz hidli moysimon suyuqliklar.
- $C_{10}$ dan boshlab (va yuqori yog‘ kislotalari: palmitin, stearin) – **qattiq moddalar**, suvda **erimaydi**.
- **Sirka kislota**: $t_{suyuq}$ = 16,6 °C – sovitilganda muz kabi qotadi (**muzsimon sirka kislota**), $t_{qayn}$ = 118 °C.
- **Chumoli kislota**: $t_{suyuq}$ = 8 °C, $t_{qayn}$ = 101 °C; teriga tushsa kuydiradi.
- Sirka essensiyasi – 70–80 % li, **stol sirkasi** – 3–9 % li sirka kislota eritmasi.

**Ehtiyot bo‘ling:** konsentrlangan sirka va chumoli kislotalar **kuydiradi** va bug‘i nafas yo‘llarini zararlaydi.